The Gauge–Gravity Stratification Sub-Programme

Presentation Note 8

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Abstract

The gauge–gravity stratification sub-programme of the Cosmochrony corpus addresses a single central question: why does the same admissible spectral functional produce Einstein gravity and Yang–Mills gauge dynamics at different spectral orders? The answer replaces the classical question "what symmetry unifies gravity and gauge?" with "at what Seeley–DeWitt order does the admissible projection respond?". Starting from the joint spectral entropy functional $S_\Pi[g, A] = \frac{1}{2} \log \det' A_{g,A}$, the sub-programme derives: the Yang–Mills equations $D_\mu F^{a\mu\nu} = 0$ from the a_4 vertical variation (Q12); the coupled Einstein–Yang–Mills system from the joint variation $\delta_{g,A} S_\Pi = 0$ (Q13); the a_6 gauge–gravity cross-coupling $R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}$ (Q13, structural); the Eddington–Born–Infeld completion of the joint functional (Q13, conditional on [H-ext]); and the structural hierarchy ratio $G_\mathrm{N} g_\mathrm{YM}^2 \sim \ell_\mathrm{sp}^2 / \dim V \cdot [\log(\Lambda/\mu)]^{-1} \ll 1$ (Q13). The stratification $a_2 \to$ gravity, $a_4 \to$ gauge, $a_6 \to$ mixed is the organising principle; the hierarchy is a structural output, not a fine-tuned coincidence.

Keywords. gauge–gravity unification; spectral stratification; Seeley–DeWitt coefficients; Yang–Mills equations; Einstein–Yang–Mills system; hierarchy ratio; Born–Infeld completion; admissible spectral entropy; non-injective projection; presentation note.

Contents

1 Motivation and Central Question

The gauge–gravity stratification sub-programme answers the following question.

Central question. The spectral gravity sub-programme (Note 4) derives the Einstein tensor as the a_2 infrared-dominant response of the horizontal metric variation of $S_\Pi[g]$. The gauge structure sub-programme (Note 3) identifies the gauge group $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ and constructs the admissible principal bundle $P_{G_\Pi}(M, G_\Pi)$. *Does the same functional $S_\Pi[g, A]$, extended to include the admissible gauge connection, also produce Yang–Mills dynamics, and if so, at what spectral order?*

The answer is yes, and the order is a_4. Gravity and gauge dynamics arise from the same functional by varying in orthogonal directions — horizontal (metric) and vertical (gauge connection) — at different Seeley–DeWitt orders. The conventional question "what symmetry unifies gravity and gauge?" is thereby replaced by:

At what spectral order does the admissible projection respond?

This is the spectral stratification principle.

The sub-programme does not derive the gauge group (Note 3) or the metric (Note 2). It derives the *dynamics* of gauge and gravitational fields from the spectral structure of the single functional $S_\Pi[g, A]$, and establishes that their UV behaviours — quadratic for gravity, logarithmic for gauge — are structural consequences of the Seeley–DeWitt hierarchy.

Remark 1 (Scope: dynamics only)**.** This sub-programme closes the *dynamical* stratification: why a_2 gives gravity and a_4 gives Yang–Mills from the same functional. It does not re-derive the gauge group or the metric. The coupled field equations with explicit fermionic matter belong to Note 6 and to the open deliverables.

Remark 2 (Vocabulary note)**.** Throughout this note, *admissibility constraint* replaces the earlier relaxation vocabulary. The sub-programme works with the effective base manifold and gauge bundle already established in Notes 2–3; no new pre-geometric substrate dynamics is introduced.

2 Position in the Cosmochrony Programme

The gauge–gravity stratification sub-programme is the capstone of the bosonic spectral sector. It completes the trilogy: Gravity 3.0 (Note 4) $\to$ Q12 $\to$ Q13.

Inputs:

- Spectral entropy functional $S_\Pi[g] = \frac{1}{2} \log \det' A_g$ and its a_2 horizontal metric variation $\to$ Einstein tensor (Note 4, Gravity 3.0 [3]).
- Admissible principal bundle $P_{G_\Pi}(M, G_\Pi)$ and gauge connection A (Note 3, Q6a [2]).
- Gauge group $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ (Note 3).
- Effective Lorentzian metric $g^{\mu\nu} = 2\eta^{\mu\nu}$ (Note 2, Q5b–Q11).
- BI saturation constant c_χ and spectral length scale ℓ_{sp} (Branch I, BornInfeld [1]).
- Horizontal–vertical decoupling of the admissible operator (Q12 [5], Lemma 1).

Outputs feed into:

- Note 6 (Fermionic Matter): Q13 is the direct predecessor of Q14; the a_4 structure provides the spectral scaffold for the projected Dirac operator.
- Phenomenological applications: gravitational-wave phenomenology, hierarchy constraints, BI mass bounds.

3 Logical Chain of the Sub-Programme

The sub-programme is organised around the spectral stratification chain:

$$\underbrace{a_2 \;\to\; G_{\mu\nu}}_{\text{horizontal } \delta_g} \;\Big|\; \underbrace{a_4 \;\to\; D_\mu F^{a\mu\nu} = 0}_{\text{vertical } \delta_A} \;\Big|\; \underbrace{a_6 \;\to\; R_{\mu\nu\rho\sigma} F^{\mu\nu} F^{\rho\sigma}}_{\text{mixed (structural)}} \tag{1}$$

from the single functional $S_\Pi[g, A] = \frac{1}{2} \log \det' A_{g,A}$.
Four conceptual stages.

1. *Extension of the operator to the gauge sector (Q12).* The Laplacian $A_g = -\nabla_g^2$ is extended to $A_{g,A} = -(\nabla^A)^2 + E$ on the associated vector bundle of $P_{G_\Pi}(M, G_\Pi)$, where E is the fibre endomorphism encoding the gauge sector curvature. The horizontal–vertical decoupling Lemma 1 of Q12 guarantees that the a_2 Einstein and a_4 Yang–Mills sectors remain independent at their respective orders.

2. *Yang–Mills from vertical a_4 variation (Q12).* The a_4 coefficient of $A_{g,A}$ contains the Yang–Mills density: $a_4(A_{g,A})|_{\text{gauge}} = \frac{1}{16\pi^2} \int \frac{1}{12} \mathrm{Tr}(F_{\mu\nu} F^{\mu\nu}) \sqrt{g}\, d^4x$. The vertical variation $\delta_A S_\Pi = 0$ yields $D_\mu F^{a\mu\nu} = 0$.

3. *Coupled Einstein–Yang–Mills system (Q13).* The joint variation $\delta_{g,A} S_\Pi = 0$ yields:
$$G_{\mu\nu} = 8\pi G_{\mathrm{N}} T^{\mathrm{YM}}_{\mu\nu}, \quad D_\mu F^{a\mu\nu} = 0,$$
where $T^{\mathrm{YM}}_{\mu\nu}$ enters via the metric dependence of the a_4 coefficient and the coupling $8\pi G_{\mathrm{N}} = c_{\mathrm{YM}}/c_{\mathrm{EH}}$ is fixed by the Seeley–DeWitt expansion alone.

4. *Structural hierarchy (Q13).* Newton's constant $G_{\mathrm{N}}^{-1} \sim \ell_{\mathrm{sp}}^{-2}$ (quadratic UV) and the gauge coupling $g_{\mathrm{YM}}^{-2} \sim \log(\Lambda/\mu)$ (logarithmic UV) share the same cutoff ℓ_{sp}. The hierarchy ratio
$$G_{\mathrm{N}} g_{\mathrm{YM}}^2 \sim \frac{\ell_{\mathrm{sp}}^2}{\dim V} \cdot \left[\log \frac{\Lambda}{\mu}\right]^{-1} \ll 1$$
is a structural consequence of the Seeley–DeWitt expansion, not a fine-tuned input.

4 Constituent Papers

4.1 Yang–Mills from the vertical variation: Q12

Q12 [5] carries out the vertical variation of $S_\Pi[g, A]$ and derives the Yang–Mills equations.

Main theorem (Theorem 1; structural, given G_Π). Let G_Π be the admissible gauge group. Then: (i) the a_4 coefficient contains $\frac{1}{16\pi^2} \int \frac{1}{12} \mathrm{Tr}(\Omega_{\mu\nu} \Omega^{\mu\nu}) \sqrt{g}\, d^4x$ with $\Omega_{\mu\nu} = F_{\mu\nu}$ (standard heat-kernel result, unconditional); (ii) the renormalized Yang–Mills term carries logarithmic coupling $g_{\mathrm{YM}}^{-2} \sim (12 \cdot 16\pi^2)^{-1} \log(\Lambda/\mu)$; (iii) the vertical variation $\delta S_\Pi / \delta A^a_\nu = 0$ yields $D_\mu F^{a\mu\nu} = 0$ in the current-free sector.

Horizontal–vertical decoupling (Lemma 1; proved). The base–fibre structure of the admissible bundle guarantees that $\delta_g S_\Pi|_{a_2}$ and $\delta_A S_\Pi|_{a_4}$ are independent: the Einstein and Yang–Mills sectors do not mix at their respective leading orders. This is not a gauge choice but a structural consequence of the principal-bundle decomposition.

Structural hierarchy of UV behaviour (§6.3; structural). $G_{\mathrm{N}}^{-1} \sim \ell_{\mathrm{sp}}^{-2}$ (quadratic) vs. $g_{\mathrm{YM}}^{-2} \sim \log(\Lambda/\mu)$ (logarithmic) from the same cutoff ℓ_{sp}: gravity and gauge have intrinsically different UV behaviours as a consequence of their Seeley–DeWitt level, not as a tuned coincidence.

Conditionalities (§7). The a_4 derivation of Yang–Mills is unconditional given any compact gauge group G_Π. The identification $G_\Pi = \mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ inherits the status of [H-color] from Note 3: unconditional for the colour-adapted graph, conditional for the standard graph at the [H-color]$_{\text{pointwise}}$ level.

Conceptual consequence (§7.5). Gravity and gauge dynamics are separated by geometric direction (horizontal vs. vertical variation on the admissible bundle) and by Seeley–DeWitt order, not by a missing symmetry group. The difficulty of reconciling gravity with the gauge-theoretic Standard Model may reflect a difference of spectral order rather than the absence of a common symmetry group G_{unif}.

4.2 Joint Einstein–Yang–Mills system and hierarchy: Q13

Q13 [4] completes the trilogy by solving the joint variational problem and deriving four items deferred from Q12.

Joint Einstein–Yang–Mills equations (Theorem 3.2; structural). The Yang–Mills stress tensor $T^{\text{YM}}_{\mu\nu}$ enters the metric equation as the unique source term produced by the a_4 coefficient under metric variation. The coupled system is:

$$G_{\mu\nu} = 8\pi G_{\text{N}} T^{\text{YM}}_{\mu\nu}, \quad D_\mu F^{a\mu\nu} = 0,$$

with coupling $8\pi G_{\text{N}} = c_{\text{YM}}/c_{\text{EH}}$ fixed by the Seeley–DeWitt expansion.

a_6 gauge–gravity cross-coupling (Theorem 4.1; structural). The leading cross-term of $a_6(A_{g,A})$ contains $R_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$, encoding the gravitational backreaction of the gauge field at higher spectral order. This is a structural identification: the a_6 coefficient is computed but its phenomenological normalisation and downstream consequences remain open.

Eddington–Born–Infeld completion (Theorem 5.3; conditional on [H-ext]). The joint functional admits a unique Eddington–BI extension: $\mathcal{S}^{\text{EBI}} \propto \int [\sqrt{-\det(g + \ell_{\text{sp}}^2 R_{\mu\nu} + \ell_{\text{sp}}^2 F_{\mu\nu})} - \sqrt{-g}]$. Conditional on [H-ext] (admissible coherence extensivity), this is the unique admissible completion of the joint spectral sector. Its structure is forced by the same Born–Infeld saturation bound that governs the gravitational sector (Note 4).

Structural hierarchy ratio (Proposition 6.1; structural).

$$G_{\text{N}} g_{\text{YM}}^2 \sim \frac{\ell_{\text{sp}}^2}{\dim V} \cdot \left[\log \frac{\Lambda}{\mu}\right]^{-1} \ll 1$$

without fine-tuning. The smallness is doubly suppressed: quadratically in ℓ_{sp}^2 (from the a_2 gravitational sector) and logarithmically in $[\log(\Lambda/\mu)]^{-1}$ (from the a_4 gauge sector). No additional input is required beyond the spectral cutoff ℓ_{sp} and the dimension $\dim V$ of the representation.

Spectral stratification as organising principle (Conclusion of Q13; structural). The prediction $a_2 \to$ gravity, $a_4 \to$ gauge, $a_6 \to$ mixed implies that fermionic structure should appear at a dedicated spectral level carrying a Dirac-type admissible operator. This prediction is the structural motivation for Q14 (Note 6) and is identified in Q13 as the central open question for the continuation of the programme.

5 Inputs from Upstream Results

1. **Spectral entropy $S_\Pi[g]$ and a_2 Einstein sector** (Note 4, Gravity 3.0 [3]): the horizontal variation and the a_2 coefficient are established prerequisites; Q12 builds on them by extending the functional to $S_\Pi[g, A]$.

2. **Admissible principal bundle and gauge group** (Note 3, Q6a [2]): the bundle $P_{G_\Pi}(M, G_\Pi)$ and connection A define the operator $A_{g,A}$. Without Note 3, the extension from A_g to $A_{g,A}$ has no canonical definition.

Table 1: Summary of results by paper. Status: P = proved (theorem-level); S = structural; C = conditional on [H-ext].

Result	Paper	Status
$a_4 \supset \frac{1}{12}\mathrm{Tr}(F^2)$ (heat-kernel)	Q12 §4	P (standard)
$\Omega_{\mu\nu} = F_{\mu\nu}$ on admissible bundle	Q12 Def. 1	S
Horizontal–vertical decoupling	Q12 Lemma 1	P
Yang–Mills $D_\mu F^{a\mu\nu} = 0$	Q12 Thm 1	S (given G_Π)
$g_{\mathrm{YM}}^{-2} \sim \log(\Lambda/\mu)$; $G_{\mathrm{N}}^{-1} \sim \ell_{\mathrm{sp}}^{-2}$	Q12 §6	S
Coupled $G_{\mu\nu} = 8\pi G_{\mathrm{N}} T_{\mu\nu}^{\mathrm{YM}}$	Q13 Thm 3.2	S
a_6 cross-term $R_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$	Q13 Thm 4.1	S
EBI joint completion	Q13 Thm 5.3	S (cond. [H-ext])
Hierarchy $G_{\mathrm{N}} g_{\mathrm{YM}}^2 \ll 1$	Q13 Prop. 6.1	S

3. **Effective Lorentzian base manifold** (Note 2, Q5b–Q11): the four-dimensional effective spacetime is the arena for the heat-kernel expansion.
4. **BI saturation constant c_χ and spectral scale ℓ_{sp}** (Branch I): the hierarchy ratio and the EBI completion both require ℓ_{sp} as input.
5. **[H-ext] for BI completion** (Note 4, Gravity 3.0): the joint EBI completion inherits the conditionality [H-ext] established for the gravitational sector.

6 Outputs to Downstream Branches

Table 2: Principal outputs and downstream consumers.

Output	Consumer	Status
$D_\mu F^{a\mu\nu} = 0$ (current-free)	SM phenomenology	S
$G_{\mu\nu} = 8\pi G_{\mathrm{N}} T_{\mu\nu}^{\mathrm{YM}}$	Q13/GW phenomenology	S
a_6 cross-coupling identified	Future precision tests	S
EBI joint functional	Astrophysical applications	S/C
$G_{\mathrm{N}} g_{\mathrm{YM}}^2 \ll 1$ structural	Hierarchy problem	S
Spectral prediction: fermions at a_{Dirac} level	Note 6 (Q14)	S
[H-color] conditionality of SU(3) sector	See Note 3	S

7 Status of Results

7.1 Proved results

1. $a_4 \supset \frac{1}{12}\mathrm{Tr}(F_{\mu\nu}F^{\mu\nu})$ (Q12 §4): standard heat-kernel result applied to $A_{g,A}$; depends only on the elliptic structure of the operator, not on the admissibility structure.

2. *Horizontal–vertical decoupling* (Q12 Lemma 1): the Einstein sector (a_2, horizontal) and Yang–Mills sector (a_4, vertical) are independent at their respective leading orders; proved from the principal-bundle structure.

7.2 Structural results

1. $\Omega_{\mu\nu} = F_{\mu\nu}$ *on the admissible bundle* (Q12): the curvature two-form of the admissible connection is identified with the gauge field strength; structural identification, not a new derivation.

2. *Yang–Mills equations from vertical variation* (Q12 Theorem 1): $D_\mu F^{a\mu\nu} = 0$ is derived from $\delta_A S_\Pi = 0$; conditional on G_Π as input.

3. *UV hierarchy:* $G_{\mathrm{N}}^{-1} \sim \ell_{\mathrm{sp}}^{-2}$ *vs.* $g_{\mathrm{YM}}^{-2} \sim \log(\Lambda/\mu)$ (Q12 §6): distinct UV behaviours are structural consequences of Seeley–DeWitt order.

4. *Coupled Einstein–Yang–Mills system* (Q13 Theorem 3.2): the joint variational problem yields the coupled system with $8\pi G_{\mathrm{N}} = c_{\mathrm{YM}}/c_{\mathrm{EH}}$.

5. a_6 *cross-coupling identified* (Q13 Theorem 4.1): the leading gauge–gravity cross-term $R_{\mu\nu\rho\sigma}F^{\mu\nu}F^{\rho\sigma}$ is identified; phenomenological normalisation and downstream consequences remain open.

6. *Structural hierarchy ratio* (Q13 Proposition 6.1): $G_{\mathrm{N}} g_{\mathrm{YM}}^2 \sim \ell_{\mathrm{sp}}^2 / \dim V \cdot [\log(\Lambda/\mu)]^{-1} \ll 1$; derived without fine-tuning from spectral data alone.

7. *Spectral stratification principle* (Q12/Q13): $a_2 \to$ gravity, $a_4 \to$ gauge, $a_6 \to$ mixed coupling is the organising principle; the hierarchy of physical interactions is identified with the hierarchy of spectral orders, not with a symmetry-breaking pattern.

7.3 Open hypotheses

1. *[H-ext]: joint EBI completion.* Theorem 5.3 of Q13 is conditional on [H-ext] (admissible coherence extensivity). This is the same hypothesis conditioning the gravitational EBI completion in Note 4; its derivation from A1–A4 is open.

2. a_6 *phenomenological normalisation.* The a_6 cross-term is structurally identified but its normalisation coefficient and observable consequences (gravitational backreaction, running couplings) are not derived.

3. SU(3) *Yang–Mills uniqueness.* The a_4 derivation holds for any compact G_Π. The internal structure of the SU(3) sector — structure constants, covariant derivative from the colour triplet data — requires a dedicated derivation not yet present.

4. *Coupled field equations with matter currents.* The Yang–Mills equations derived here are in the current-free sector. The full equations $D_\mu F^{a\mu\nu} = J^{a\nu}$ with fermionic matter currents from Note 6 are open.

5. *[H-color] for the* SU(3) *sector.* Inherited from Notes 1 and 3: the SU(3) identification is established unconditionally on the colour-adapted graph and at three analytical levels on the standard graph. The [H-color]$_{\mathrm{pointwise}}$ component remains open.

8 Open Deliverables

Three open deliverables bound the sub-programme.

Open deliverable 1: a_6 coupling normalisation. The a_6 gauge–gravity cross-coupling is structurally identified. Deriving its normalisation coefficient would give a quantitative prediction for the leading gauge–gravity mixing at high spectral order and would constrain the EBI completion.

Open deliverable 2: Derivation of [H-ext]. [H-ext] is the sole remaining conditionality for the EBI joint completion. Its derivation from A1–A4 would promote Theorem 5.3 of Q13 to an unconditional theorem. The structural mechanism is identified (BI parity at the scalar and tensorial levels); the admissible coherence extensivity step is the analytical gap.

Open deliverable 3: Full coupled equations with matter. The primary missing piece is the coupled system $G_{\mu\nu} = 8\pi G_{\mathrm{N}}(T^{\mathrm{YM}}_{\mu\nu} + T^{\mathrm{ferm}}_{\mu\nu})$, $D_\mu F^{a\mu\nu} = J^{a\nu}$ with explicit fermionic matter from Note 6. This requires integrating the projected Dirac sector of Q14 into the joint variational framework of Q13.

9 Conclusion

The gauge–gravity stratification sub-programme establishes that gravity and Yang–Mills gauge dynamics arise from the same admissible spectral functional $S_\Pi[g, A]$, separated by geometric direction and Seeley–DeWitt order.

The Yang–Mills equations follow from the a_4 vertical variation (Q12, structural given G_Π). The coupled Einstein–Yang–Mills system follows from the joint variation (Q13, structural). The a_6 gauge–gravity cross-coupling is structurally identified (Q13, structural). The hierarchy ratio $G_{\mathrm{N}} g^2_{\mathrm{YM}} \ll 1$ is a structural consequence of the spectral stratification, requiring no fine-tuning (Q13, structural). The EBI completion of the joint functional is the unique admissible UV completion, conditional on [H-ext] (Q13).

The spectral stratification $a_2 \to$ gravity, $a_4 \to$ gauge, $a_6 \to$ mixed is an organising principle rather than a theorem: it is a structural pattern identified by the Seeley–DeWitt level at which the same admissible operator responds to variation. Its prediction that fermionic structure should appear at a dedicated Dirac-type spectral level is the structural motivation for the fermionic matter sub-programme (Note 6, Q14).

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